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INSTITUTO
UNIVERSITÁRIO
DE LISBOA

Dissertação em Engenharia

Types of Dissertation Projects

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Outline

- 1. Dissertation and Research**
- 2. Types of Research**
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- 5. Presenting Non-Numerical Data**
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1. Dissertation and Research

- **A thesis project**
 - builds and tests the skills and knowledge acquired during your education and training in previous courses
 - allows training in carrying out projects independently, at an advanced level, using a sound method.
 - deepens your understanding of a subject → research and contribute to the common knowledge
- Research → diligent and systematic inquiry or investigation in an area, with the objective of discovering or revising facts, theories, applications etc.

2. Types of Research

- **Two approaches of research: Qualitative Research and Quantitative Research.**
- **Qualitative Research:**
 - deep origin in social science
 - Involves the use of qualitative data
 - interview, documents, observation, questionnaire and the researcher's impression and perception
 - Can serve one or more purposes:
 - Description, Interpretation, Verification, Evaluation
 - Forms of Qualitative Research:
 - Phenomenological Method (deriving from phenomena)
 - Ethnographic Model.
 - Grounded Theory Method.
 - Case Study Model.
 - Historical Model.
 - Narrative Model.

2. Types of Research

- **Quantitative Research:**

- origin in the natural sciences
- systematic investigation of phenomena by gathering numerical data and performing statistical, mathematical, or computational techniques.
- Purposes:
 - Statistical model, objectivity, replication, generalization, cause and effect relationships.
- Types of Quantitative Research:
 - Descriptive Research
 - Correlational Research
 - Experimental Class Research
 - Comparative Research

2. Types of Research

- It is the nature of the problem or phenomenon itself, which guides the decision as to which method to use.
- You choose and use tools once
 - you have established the nature of the problem,
 - and when you know the hypothesis/proposition that should be tested.
 - choice of methods to use should be discussed with supervisor.

3. Dissertation Projects

- **Descriptive Projects**

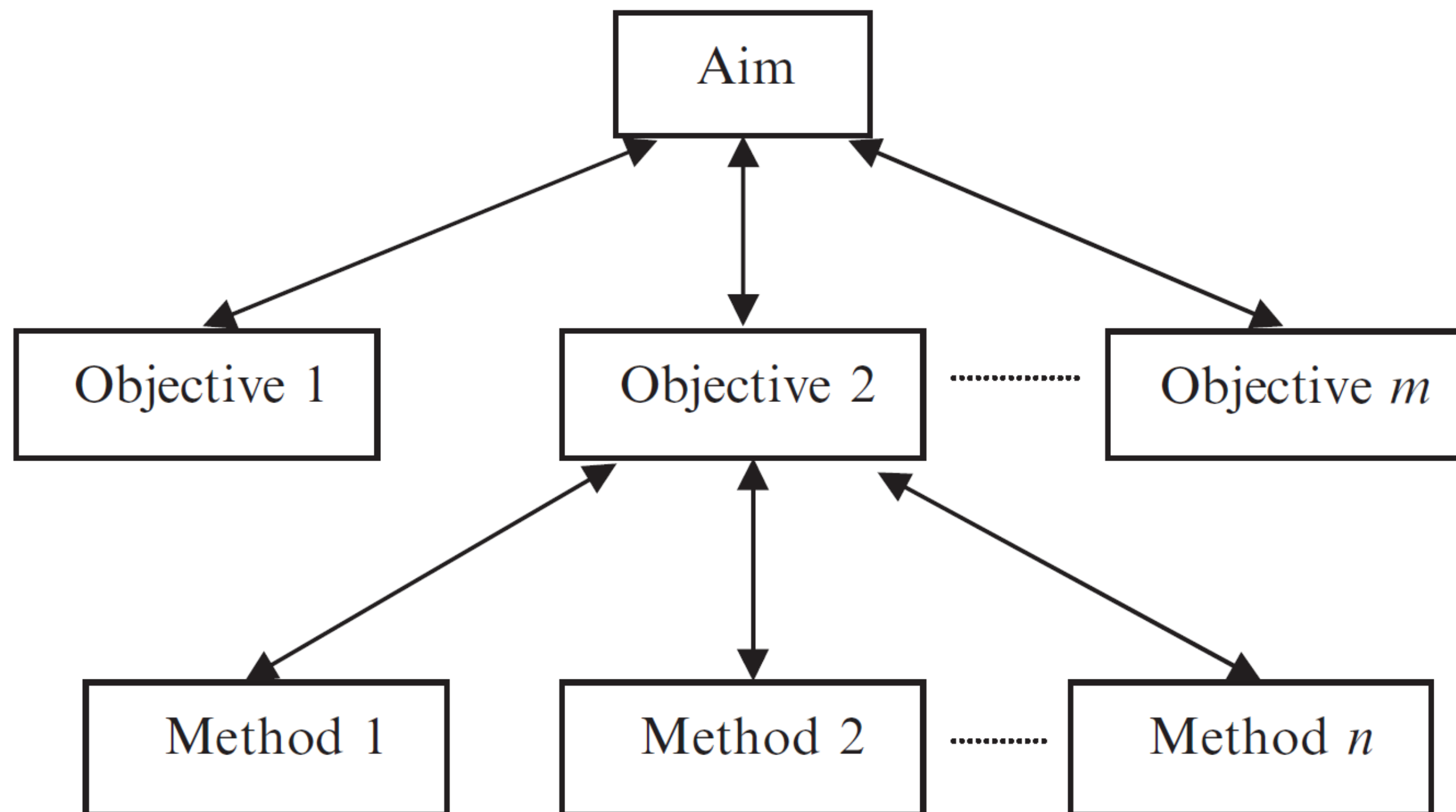
- present the state-of-the art for a given subject
- can be aimed at categorizing and comparing previous work within a subject area.
 - identify how a subject area has evolved over time, what its current status is, and how it may evolve in the near future.
- can also be aimed at gaining an understanding of the current status of the subject, and to identify important factors.
- Descriptive projects should not be simply a summary of the literature → analysis should be provided

3. Dissertation Projects

- **Theory Oriented Projects**
 - deal with extending or comparing existing theoretical models without testing them in practice
 - important to be aware of how the theoretical ideas may be applied in practice even if no implementation is done.
- **Applied Projects**
 - deal with conducting experiments and building proof-of principle implementations.
- **Comparison of Theory and Practice Projects**
 - Projects which combine theory and practice may contrast the theory with current practice in companies or organizations.
- **Most thesis projects use elements from more than one of the above categories**

4. Developing the objectives

- After defining the project aim and objectives it is necessary to choose a method for each objective.



4. Developing the objectives

- Simple analogy of changing the color of a car from green to red:

- Aim

- “change colour of car to red”

- Objectives/sub-goals:

- ✓ Remove parts that should not be painted
 - ✓ Buy paint
 - ✓ Paint car
 - ✓ Re-assemble removed parts once the paint is dry

- Possible methods for objective “Paint car”

- 1. Using an ordinary paintbrush
 - 2. Using a roller
 - 3. Using an airbrush tool
 - 4. Paying somebody to do the job

4. Developing the objectives

- **Methods should be chosen considering validity and reliability.**
- **Examples of methods/techniques:**
 - Literature Analysis
 - systematic examination of a problem, by means of an analysis of published sources, undertaken with a specific purpose in mind
 - not to be confused with a usual review of existing work in an area for familiarization
 - Interview
 - can be undertaken in a variety of different ways, and to different ends
 - open interviews are commonly used in qualitative research → specific issues to be covered are not completely planned in advance.
 - closed/pre-structured interview is characterized by a fixed set of questions → more suitable when using statistical methods to analyze the results

4. Developing the objectives

➤ Case Study

- in-depth exploration of a phenomenon in its natural setting.
- Important to consider the extent to which the findings can be generalized.

➤ Survey

- often used for exploring a relatively well-known phenomenon
- closely associated with the use of questionnaires, and statistical techniques for analysis

➤ Implementation

- Many projects in engineering consist of developing new solutions
- Necessary to implement the proposed solution, to demonstrate its advantages.
 - demonstration depends largely on what type of solution you are proposing
- Needs to be compared with implementations of existing solutions, before conclusions can be drawn

4. Developing the objectives

➤ Experiment

- Typically used to verify or falsify a previously formulated hypothesis.
 - the hypothesis is usually formulated as a causal relationship
- There is a large number of possible things that may go wrong in any experiment, and which may lead to erroneous and misleading results.
- A positive outcome lends some support to the hypothesis, but usually we cannot claim that it “proves” its correctness

5. Presenting Non-Numerical Data

➤ Literature Analysis

- a literature analysis can be structured in such a way that it provides an argument for a certain idea or interpretation.
- there are many different ways a literature analysis can be structured →no “fixed” format to follow.
- can learn through examples.

5. Presenting Non-Numerical Data

➤ Interviews and Questionnaires

- material must now be structured and presented to the reader in an appropriate form
- data from questionnaires can often be presented using figures or tables
- different examples of structuring data from interviews

	No grouping	Grouping	Summary
Structure by company	Company A • Q1 • Q2 • Qn Company B • Q1 • Q2 • Qn	Company A • Q1-Q5 • Q6-Q8 Company B • Q1-Q5	Company A • Summary of Q1-Qn Company B • Summary of Q1-Qn
Structure by question	Q1 • Company A • Company B Q2 • Company A • Company B Qn • Company A • Company B	Q1-Q5 • Company A • Company B Q9-Qn • Company A • Company B	Q1-Q5 • Summary of company A and B Q9-Qn • Summary of company A and B

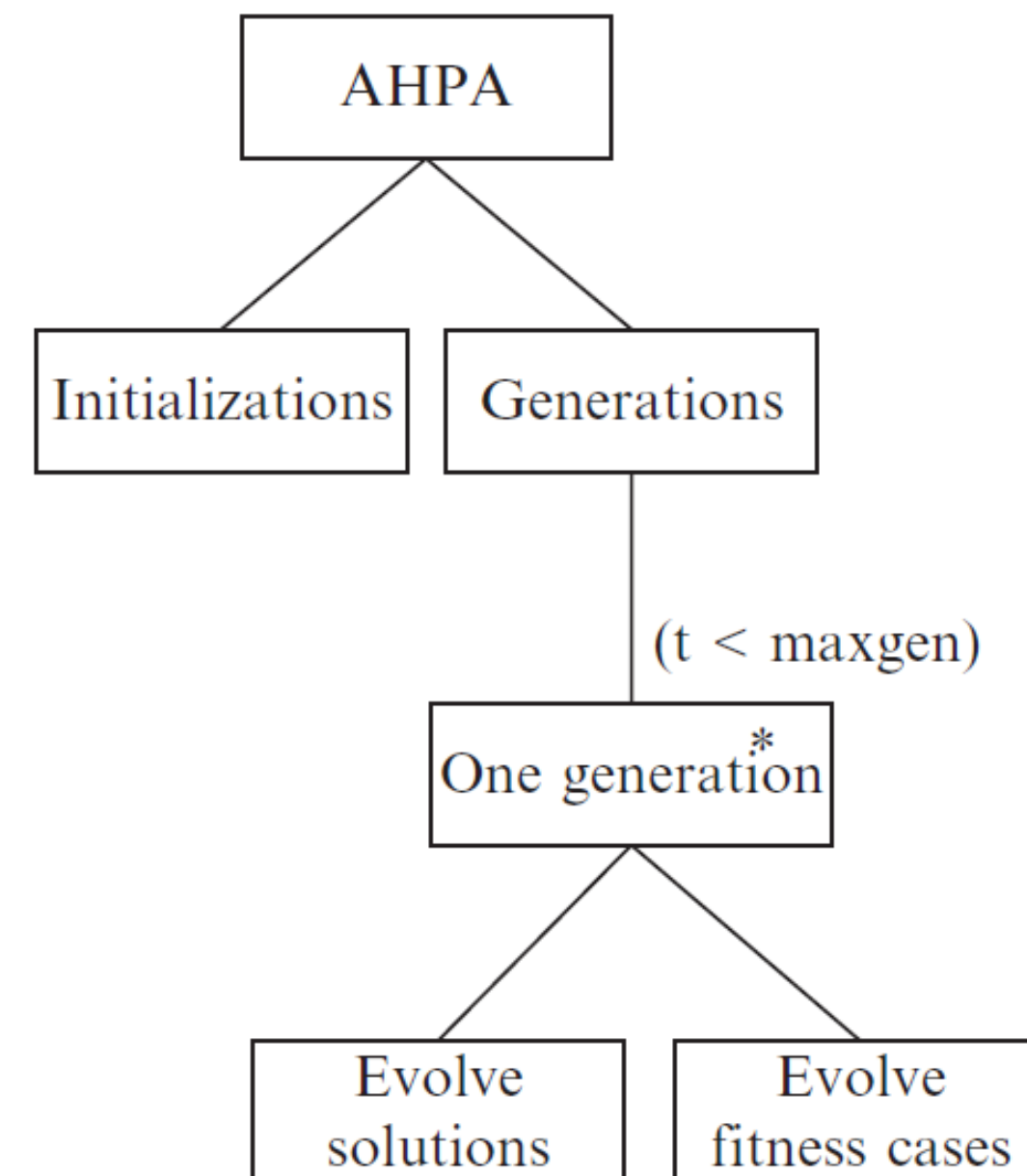
5. Presenting Non-Numerical Data

➤ Implementation

- Implementation should be well documented in order to convince the reader that it is correct.
- show the steps in the process of developing the implementation
 - pseudo-code, graphical representations (flowcharts, block-diagrams,...), etc.

```

initialize(  $H^e$  )
initialize(  $H^s$  )
initialize(  $P^e$  )
initialize(  $P^s$  )
while (  $t < maxgen$  ) do
    until ( (  $h \in H^e$  ) and (  $h$  solves  $P^e \cup (P^s)$  ) )
        evaluate(  $H^e, P^e \cup P^s$  )
        reproduce(  $H^e$  )
         $t = t + 1$ 
    end
    add  $h$  to  $H^s$ 
    until ( (  $p \in P^e$  and (  $p$  defeats  $H^e \cup H^s$  ) ) )
        evaluate(  $P^e, H^e \cup H^s$  )
        reproduce(  $P^e$  )
         $t = t + 1$ 
    end
    add  $p$  to  $P^s$ 
end
    
```



5. Presenting Non-Numerical Data

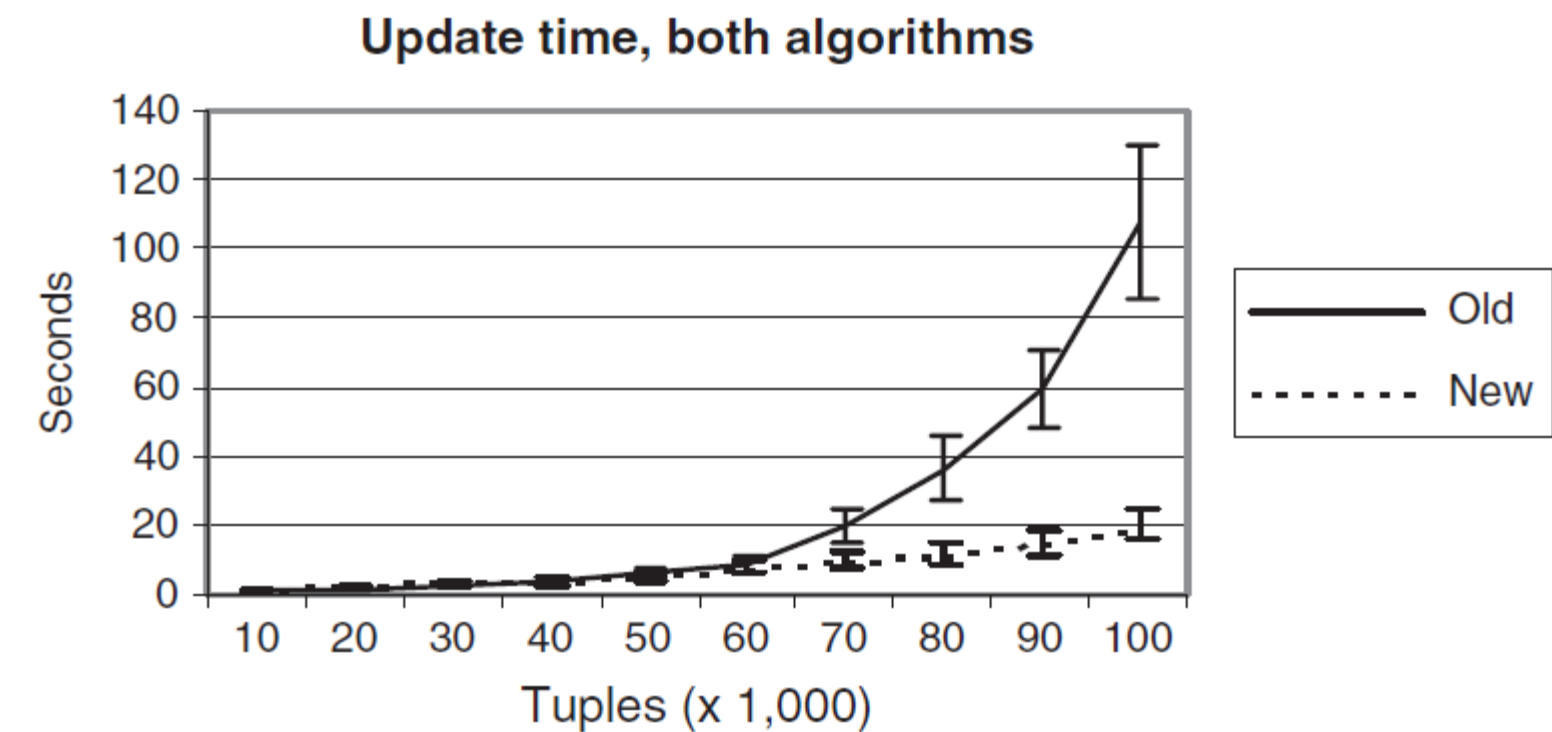
➤ Implementation

- Most of the code should be placed as an appendix.
- Depending on the purpose of the implementation, other information can be provided: hardware and software requirements, user instructions etc.
- When implementing a simulator, it should be validated by using reference algorithms

6. Presenting Numerical Data

➤ Resort to Tables and Graphs

Tuples (× 1,000)	Update time (seconds)	
	Old algorithm	New algorithm
10	0.8	1.1
20	1.6	2.1
30	2.5	3.2
40	3.3	3.9
50	6.2	5.1
60	8.9	7.5
70	19.6	9.9
80	36.4	11.3
90	59.7	15.4
100	108.2	20.3



- if the studied system is stochastic, experiment/simulation run must be repeated several times
- a suitable test for statistical significance is necessary before attempting to draw any conclusions

7. Dissertation Examples

➤ Case-Study



Instituto Universitário de Lisboa

Department of Information Science and Technology

Control of Robot Swarms Through Natural Language Dialogue

A Case Study on Monitoring Fires

Nuno Miguel Amorim Mendonça

A Dissertation presented in partial fulfillment of the Requirements
for the Degree of

Master in Computer Engineering

Supervisor

Sancho Moura Oliveira, Assistant Professor, Ph.D.

ISCTE-IUL

Co-Supervisor

Ricardo Daniel Santos Faro Marques Ribeiro, Assistant Professor,
Ph.D.

ISCTE-IUL

October, 2019

7. Dissertation Examples

➤ Literature Review (Multivocal Literature Review - MLR)

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Case study analysis of the relationship between capabilities, practices, challenges, and benefits when employing DevOps

Alfredo Tiago Fânzeres Nunes Martins

Department of Information Sciences and Technologies

Master's in Telecommunications and Informatics Engineering

Orientador(a):
Phd. Rúben Filipe de Sousa Pereira, Professor Assistente
ISCTE-IUL

Co-Orientador(a):
Phd. João Carlos Marques Silva, Professor Assistente
ISCTE-IUL

October, 2022

7. Dissertation Examples

➤ Entrevistas+Implementação+Literature Review (MLR)

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A Mobile CRM Development for Real Estate

João Pedro Beato Antão

Master in Telecommunications and Computer Engineering

Supervisor:
Prof. Dr. Rúben Filipe de Sousa Pereira, Assistant Professor,
Iscte – Instituto Universitário de Lisboa

Co-supervisor:
Prof. Dr. Ricardo Daniel Santos Faro Marques Ribeiro, Associate Professor,
Iscte – Instituto Universitário de Lisboa

november, 2020

7. Dissertation Examples

➤ Implementação + Experimental

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Molecular Communications techniques for the Internet of Bio-Nano Things

Sofia Alexandra Duarte de Figueiredo

Master Degree in Telecommunications and Computer Engineering

Supervisor

PhD Nuno Manuel Branco Souto, Associate Professor (with Habilitation)

Iscte-IUL

Co-Supervisor

PhD Francisco António Bucho Cercas, Full Professor

Iscte-IUL

November, 2021

7. Dissertation Examples

➤ Implementação (Ferramenta Software) + Experimental

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Avaliador Pedagógico da Qualidade de Código

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Mestrado em Engenharia de Telecomunicações e Informática

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Outubro, 2020

Sources:

Berndtsson, M., Hansson, J., Olsson, B., Lundell, B. "Thesis Projects: a Guide for Students in Computer Science and Information Systems." Springer, London (2008).

Kothari, C. "Research methodology: methods and techniques," New Age International, New Delhi, India (2004)